Michael Plesser

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Education

Pennsylvania State University (PSU) — PhD in Physics **2022 - 2027**
John Randall Shuman Troxell Memorial Scholarship in Physics 2025
Edward A. and Rosemary A. Mebus Graduate Fellowship in Physics 2024
Edward M. Frymoyer Honors Scholarship 2024

Pennsylvania State University (PSU) — MSc in Physics **2020 - 2022**
Paul Berg and Daniel J. Larson Distinguished Graduate Fellowship 2020

The University of Hong Kong (HKU) — Semester Abroad **2016**

Northeastern University (NEU) — BSc in Physics with Honors **2015 - 2019**
Merit scholarship for four years, also awarded a Dean's scholarship
Presidential global scholarship to perform research in Italy
Presidential global scholarship to study in Hong Kong

Research

PSU — PhD Supervisor: Dr Jacob Bourjaily 2022 - 2027

Studying mathematical structures and "on-shell" methods in QFT. Ongoing work includes…
Colour in gauge theory, applications of classical and quantum representation theory,
Covariant formalisms for massive particles and generalized recursion relations,
Mathematica packages for handling the math and physics of scattering,
Generalized double copy and colour-kinematics duality,
Combinatorics of amplitudes.

NEU — Supervisor: Dr James Halverson 2019 - 2020
Studied solvability of Diophantines in string index calculations under topological transitions.

CERN / CMS / NEU — Supervisor: Dr Toyoko Orimoto 2018 - 2019
Used ROOT to perform test beam timing resolution analyses on HL-LHC ECAL hardware.

Università degli Studi di Milano — Supervisor: Dr Marcel Di Vece 2017 - 2018
Modeled "planetary" nanoparticle systems formed by a sputtering cluster source in Python.

CERN / ATLAS / HKU — Supervisor: Dr Yanjun Tu 2016 - 2017
Used ROOT to simulate backgrounds and search for SUSY in dilepton events at CERN.

Publications and Preprints

1. S. Caron-Huot, J. Fei, **M. Plesser**, *Canonical Bases for Classical Group Tensor Products,* To appear: 2026
2. J. Bourjaily, **M. Plesser**, *Covariant BCFW for all Masses***,** *2610.10686*
3. J. Bourjaily, **M. Plesser**, P. Velie, *Constructing Concrete Colour Tensors for Simple Lie Algebras,* 2609.13365
4. J. Bourjaily, **M. Plesser**, P. Velie, *Tree Amplitudes with Charged Matter in Pure Gauge Theory*, 2604.18672
5. J. Bourjaily, **M. Plesser**, C. Vergu, *The Colour Dependence of Amplitudes*, **JHEP** 08 (2026), 2512.23806
6. J. Bourjaily, **M. Plesser**, C. Vergu, *The Many Colours of Amplitudes***,** 2412.21189
7. J. Bourjaily, **M. Plesser**, et al, *Gauge-Invariant Double-Copies via Recursion Relations,* **Phys. Rev. Lett.** 131 (2023), 2307.02542
8. J. Halverson, **M. Plesser**, F. Ruehle, J. Tian, *Kahler Moduli Stabilization and the Propagation of Decidability*, **Phys. Rev. D** 101 (2020), 1911.07835
9. S. Marom, **M. Plesser**, et al, *`Planetary' silver nanoparticles originating from a magnetron sputter plasma*, **J. Phys. D: Appl. Phys** 52 (2019)

Invited Talks and Visits

Invited talk at Penn State's Math department PSU, PA, USA, 2026

"Canonical bases for quantum groups and applications to physics"

Invited to visit McGill University by Simon Caron-Huot *(2 weeks)* McGill, QC, CA, 2026

"Colour in Gauge Theory: why would I Lie to you?"

Invited to visit the IAS by Jacob Bourjaily *(1 week)* IAS, NJ, USA, 2026

Invited to visit the IAS by Nick Early, Nima Arkani-Hamed *(1 week)* IAS, NJ, USA, 2026

Invited to visit UCT by Jeff Murugan and Jaco Van Zyl *(1 week)* UCT, ZA, 2025

"What can representation theory teach us about gauge theory?"

Invited talk at April APS Conference Denver, CO, USA 2019

"Test beam studies of precision timing with the upgraded CMS-ECAL"

Conferences, Summer Schools, and Workshops

AI in CAmplitudes UCLA, CA, USA 2026
Loop-The-Loop 3 Workshop on Amplitudes (Online), 2026
Workshop on N=4 at the IAS hosted by Nima Arkani-Hamed IAS, NJ, USA, 2026
Prospects in Theoretical Physics (PiTP) Summer School IAS, NJ, USA, 2026
Gong show: "QED from N=4"
Amplitudes Summer School U. of Southampton, UK, 2026
Symbology@15 Conference MPP, Munich, DE, 2025
Gong show: "Masses, flavours, and colours in amplitudes"
TASI Summer School CU Boulder, CO, USA, 2025
Gong show: "Colour structures and amplitudes"
Michigan Interdisciplinary Meeting on Amplitudes U. of Michigan, MI, USA, 2025
Gong show: "Graph games from representation theory"
Loop-The-Loop 2 Workshop on Amplitudes (Online), 2025
Strings and Geometry Conference ICTP Trieste, IT, 2025
Loop-The-Loop Workshop on Amplitudes (Online), 2024
Mathematical Aspects of Scattering Amplitudes Conference GGI, Florence, IT, 2024
Amplitudes Conference and Summer School IAS, NJ, USA, 2024
Gong show: "A ridiculous formula for the number of SU(N) gluonic colour factors"
Amplitudes Conference and Summer School CERN, FR / CH, 2023
Gong show: "Covariant BCFW for all masses"
Amplitudes Conference and Summer School Charles University, Prague, CZ, 2022
The Amplitude Games Summer School MITP (Online), 2021
Quantum Gravity Summer School Bard College, NY, USA, 2019

Teaching Assistantships at Penn State University

Introductory Mechanics *(PHYS 211)* 2024, 2026
Introductory Electromagnetism *(PHYS 212)* 2022, 2023, 2024
Thermal Physics *(PHYS 213)* 2022
Wave Physics *(PHYS 214)* 2022, 2025
Algebra-Based Introduction to Mechanics *(PHYS 250)* 2025
Electromagnetism *(PHYS 400)* 2021
Subatomic Physics *(PHYS 406)* 2022, 2025
Quantum Mechanics I *(PHYS 410)* 2021, 2025
Graduate Quantum Mechanics II *(PHYS 562)* 2022
Computational Resources and Curriculum Development 2026

Outreach and Mentorship

1) Provided tutoring and career advice to minority students at the HBCU (Historically Black College or University) NCCU (North Carolina Central University) as part of the PREM (Partnership for Research and Education in Materials) program

Graduate Electromagnetism	2024, 2025
Undergrad Electromagnetism and Quantum Mechanics	2023, 2024, 2025

2) Peer-mentor for junior PhD students working in Dr. Bourjaily's group at Penn State

Philip Velie	2023-2027
Malek Telfah	2026-2027

3) Tutored Penn State physics PhD students remediating first-year courses 2025
4) Research talks at Penn State's "Primordial Universe and Gravity" (PUG) seminar

"Why Do We Care About N=4 Super Yang-Mills?"	2026
"Why (Non-Abelian) Gauge Theory?"	2025
"Correcting Coulomb and Newton with On-Shell Methods"	2024
"Quantum Field Theory Without Fear"	2023

5) Research talk to the physics and chemistry club at State Area High School

"A Brief History of QFT and Why the Sky is Blue"	2022

6) Research talk at Penn State's "Mathematical Aspects of Physics" (MAP) seminar

"Why Good Variables Matter: Representations of Poincaré"	2022

Rock Climbing

- Semi-professionally sponsored by Mad Rock Climbing from 2014 - present.
- Commercial routesetter with over a decade of experience.

    - *Routes:*
      Salathé Wall on **El Capitan** (5.9 C2, 3500' Yosemite, USA),
      Snake Dike on **Half Dome** (5.7 R, 2000' Yosemite, USA),
      Time Wave Zero on **El Toro** (5.12a, 2300' El Potrero Chico, Mexico),
      Flatirons scramble during the TASI 2025 summer school in Boulder, CO
      El Camino Royale 500' - **Royal Arch** 100' - **Dodge Block** 700'

    - *Boulders:*
      Mooiste Meisie (v13 / 8B, South Africa),
      Freak Brothers (v11 / 8A, Switzerland),
      Monstrous Moonshine (v11 / 8A FA, NC, USA),
      973 to the Dangerzone (v6 / 7A R FA, Hong Kong)